SOFTWARE ENGINEERING MINI-PROJECT

Software Architecture & Design Specification

Real-Time Chat Application Using Socket Programming

Deliverable 3 – IEEE-Style Architecture and Design

Document Control

Item	Details
Document	Software Architecture & Design Specification
Project	Real-Time Chat Application Using Socket Programming
Version	1.0
Architecture Style	Client–Server with layered responsibilities

1. Introduction

1.1 Purpose

This document defines the software architecture and detailed design of the Real-Time Chat Application. It describes the major components, architectural pattern, security architecture, interfaces, sequences, error handling, and traceability to the SRS.

1.2 Scope

The design covers the client, chat server, authentication, connection/session management, message routing, validation/security, user data, and communication interfaces.

1.3 Design Goals

- Real-time bidirectional communication.
- Clear separation between client and server responsibilities.
- Support for multiple concurrent users.
- Input validation and access control.
- Maintainable and testable components.

2. Architecture

2.1 Architectural Pattern

The system uses a Client–Server architecture with layered responsibilities. Clients initiate communication with the server, while the server manages authentication, active connections, message routing, and access control. This separation supports independent development and testing of client and server functionality.

2.2 Component Diagram

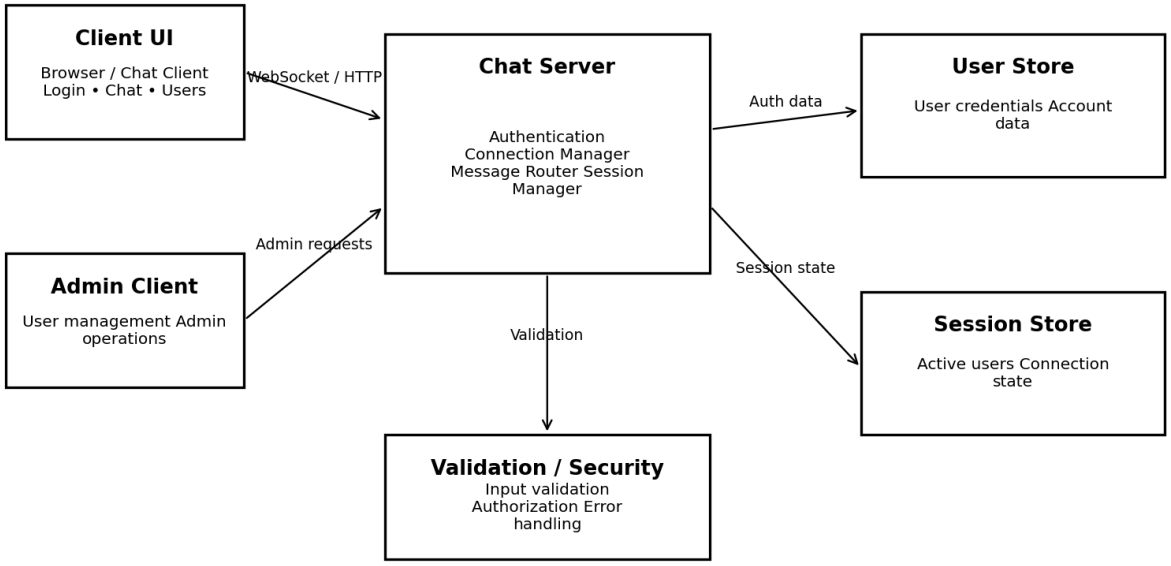


Figure 1. High-level component architecture of the chat application.

2.3 Component Description

Component	Responsibility	Related SRS
Client UI	Registration/login, chat interface, messages, timestamps, online users, logout.	FR-01–FR-04, FR-06–FR-18
Chat Server	Accept connections, authenticate users, maintain sessions, route messages.	FR-02, FR-05, FR-08, FR-12–FR-20
Connection Manager	Create/maintain/terminate socket connections and connection status.	FR-05, FR-13–FR-15, FR-20
Message Router	Routes valid messages to intended recipient(s).	FR-07–FR-09, FR-19
Validation/Security	Input validation, authentication and authorization checks.	FR-17, FR-18, SEC-01–SEC-04, NFR-09
User Store	Stores/retrieves account information required for authentication.	FR-01–FR-03
Session Store	Maintains active-user and connection state.	FR-13, FR-16

2.4 Data and Communication Flow

1. Client sends an authentication or chat request to the server.
2. Server validates the request and checks authorization where required.
3. For a valid message, the server routes the message to the appropriate recipient(s).
4. The client displays the received message and relevant status information.
5. Connection/session state is updated when users join, leave, or disconnect.

3. Traceability to Requirements

Architecture/Design Element	SRS Trace	Purpose

Authentication component	FR-01–FR-03, SEC-01	User registration and access control
Connection Manager	FR-05, FR-13–FR-15, FR-20	Persistent connections and disconnection handling
Message Router	FR-07–FR-09, FR-19	Real-time message delivery
User list/session state	FR-12, FR-16, NFR-02	Concurrent users and active-user visibility
Validation/Security layer	FR-17, FR-18, NFR-09, SEC-02	Input validation and security
Server modularity	NFR-06, NFR-08	Maintainability and scalability

4. Security Architecture

4.1 Security Controls

- Authentication is performed before protected chat operations are permitted.
- Authorization checks are applied to administrative operations.
- User-provided data is validated before processing.
- Unauthenticated clients are prevented from accessing protected messaging functions.
- Session/connection state is controlled by the server.

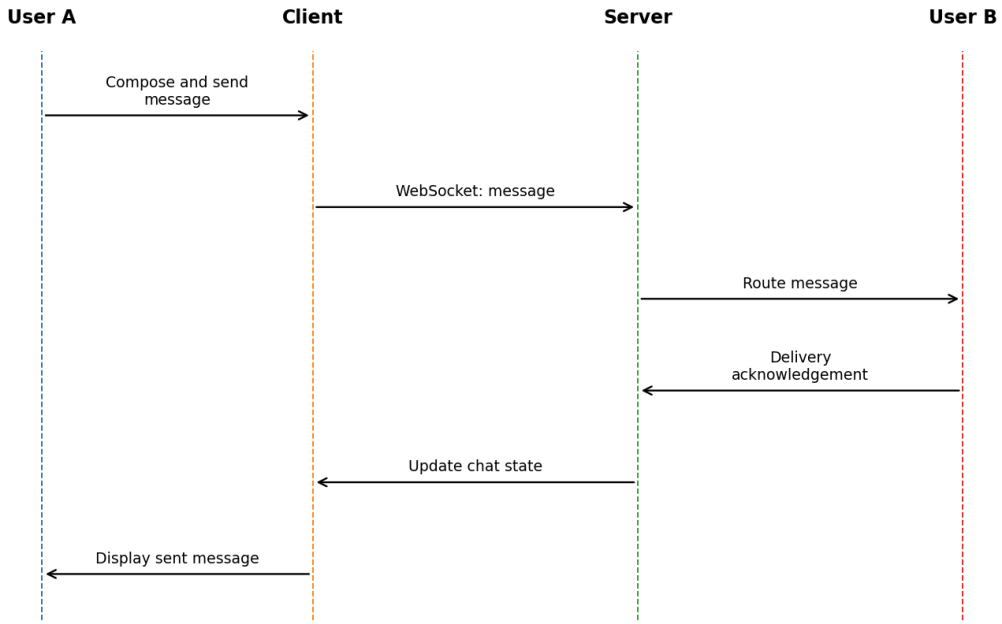
4.2 Security Objectives

6. Prevent unauthorized access to protected chat functionality.
7. Maintain integrity of server-side operations through validation and access control.

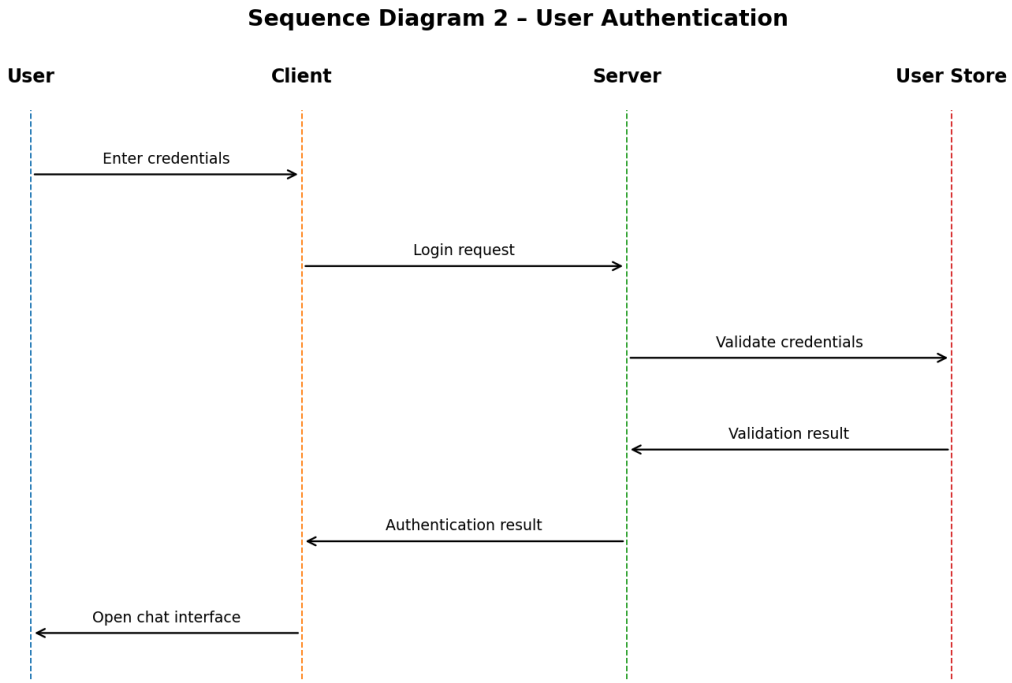
5. Detailed Design

5.1 Sequence Diagram – Sending a Message

Sequence Diagram 1 – Sending and Receiving a Chat Message



5.2 Sequence Diagram – Authentication



5.3 API / Communication Design

Operation	Direction	Input	Output	Purpose
REGISTER	Client → Server	Username + registration data	Success/error response	Create account
LOGIN	Client → Server	Credentials	Authentication result/session	Authenticate user
JOIN_CHAT	Client → Server	Authenticated session	Join/status response	Enter chat service
SEND_MESSAGE	Client → Server	Recipient/chat + message	Acknowledgement/error	Send message
MESSAGE	Server → Client	Routed message + timestamp	Display/update	Deliver message
USER_LIST	Server → Client	Active session state	Connected-user list	Show online users
LOGOUT	Client → Server	Authenticated session	Logout/status	End session

5.4 Error Handling

Error Condition	System Response	Related SRS
Invalid credentials	Reject login and display an authentication error.	FR-03
Empty/invalid message	Reject message before transmission.	FR-17, FR-18
Server unavailable	Notify user that connection cannot be established.	FR-20
Unexpected disconnect	Update connection state and notify affected user(s) as applicable.	FR-15, FR-20
Unauthorized operation	Reject operation and return an access-control error.	SEC-01, SEC-04
Malformed input	Reject or safely handle input without processing it as valid data.	NFR-09, SEC-02

6. Deployment Considerations

For the student implementation, the server may run on localhost or a controlled network machine. Clients connect using the configured server address and communication port. The selected runtime and dependencies must be installed before deployment.

7. Design Constraints and Assumptions

- The selected socket/WebSocket framework remains the primary communication mechanism.
- The initial design focuses on text messaging.
- Database/session persistence details may be adapted to the selected implementation.
- The architecture is intended to support the requirements defined in the SRS.